

The Re-prompt Tax Persists: Measuring Hidden Interaction Costs for Multilingual and Code-Switched LLM Users

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Abstract

Multilingual users often interact with large language models (LLMs) through code-switched prompts, mixed-language editing requests, and constraints about script, register, locale, and literal text preservation. Standard one-turn evaluations usually ask whether the final answer is correct, but they miss a common usability failure: the model returns a fluent, plausible answer that violates the user’s interaction contract, forcing the user to spend extra turns repairing the response. We introduce *Re-prompt Tax*, a metric family that measures first-turn global alignment, the number of repair turns needed for recovery, unresolved failures, and token overhead. We construct REPROMPTTAX-STRESS-V0.2, a 120-item synthetic stress pilot across Spanish-English, Hindi-English, and Arabic-English interactions, and evaluate three GPT-4.1-family API models under a baseline prompt and a Global Interaction Contract prompt. Baseline first-turn global alignment ranges from 67.5% to 76.7%; the mitigation improves alignment to 76.7–93.3% and reduces mean repair turns and repair-normalized token tax. A current-model refresh shows the same pattern on full 120-item gpt-5.4-mini and gpt-5.5 runs: under gpt-5.5, alignment rises from 81.7% to 98.3%. Prompt-mechanism diagnostics show that a narrower content-preservation scaffold captures most of this gain, but only near-ties the full contract on current models. Analysis shows that the dominant failure mode is not generic multilingual QA, but implicit editing preservation: models often translate or frame content in the instruction language when the user intended the quoted content to remain in English. A 10% stratified blinded LLM-judge audit agrees with the automatic scorer on 71/72 sampled pass/fail labels, and a gpt-5.5 judge refresh agrees on 70/72. We release a conservative evaluation protocol and claim boundary for measuring hidden global usability costs rather than broad multilingual ability.

1 Introduction

The GenAI4World workshop asks how generative AI systems should be evaluated for global tasks, contexts, vulnerabilities, and usability beyond standard accuracy metrics (GenAI4World Organizers, 2026). This question matters because many real users do not interact with LLMs through clean monolingual benchmark items. A single request may contain an instruction in one language, content in another, an English file name embedded in Hindi written in Latin script, or a local date, currency, or honorific convention. A model may understand the literal task and still violate the pragmatic contract that makes the response usable.

Consider the following Spanish-English editing request:

Me ayudas a mejorar este texto para que suene natural? “I can’t join the call today, but I can share my update by email.”

A globally aligned assistant should infer that Spanish is the instruction language and that the quoted English sentence is the object of editing. Unless translation is requested, the edited sentence should remain English. A response that translates the sentence into Spanish is

43 fluent and superficially helpful, but it is unusable for the user’s intended workflow. The user
 44 must diagnose the failure, issue a corrective follow-up such as “Please keep the rewritten
 45 text in English,” and check whether the repair succeeded. We call this extra burden the
 46 *Re-prompt Tax*.

47 Existing multilingual evaluations have greatly expanded language coverage and have
 48 shown that translated benchmarks can be culturally distorted (Singh et al., 2024; Xuan et al.,
 49 2025; Huang et al., 2025b). Other work has studied multilingual instruction following (Zhou
 50 et al., 2023; Li et al., 2025b) and output-language alignment in code-switched interactions
 51 (Oh et al., 2026). These efforts are essential, but they leave an interactional gap. A user-facing
 52 system is not only a question-answering engine; it is a collaborator in a multi-turn workflow.
 53 When a first response violates a language, script, preservation, register, or locale constraint,
 54 the cost is paid by the user in extra turns, extra tokens, and uncertainty about whether the
 55 model will repair the failure.

56 This paper treats multilingual usability as an *interaction contract* between a user and a model.
 57 The contract may specify, explicitly or implicitly: (1) the task, (2) the expected response
 58 language, (3) the language of content to be edited or summarized, (4) script or romanization
 59 requirements, (5) literal spans that must not be changed, and (6) register or locale constraints.
 60 A model can violate this contract even when a semantic task label such as “rewrite” or
 61 “summarize” is satisfied. Measuring only one-turn task success therefore understates the
 62 burden imposed on multilingual and code-switched users.

63 Our contributions are:

- 64 • We define **Re-prompt Tax**, a metric family for hidden multilingual repair burden:
 65 first-turn global alignment, re-prompt turn tax, Repair@1/2, unresolved rate, and
 66 repair-normalized token tax.
- 67 • We construct REPROMPTTAX-STRESS-V0.2, a balanced 120-item synthetic stress
 68 pilot over three language pairs and four task families: implicit editing preserva-
 69 tion, output-language inference, quote preservation, and script/register/locale
 70 constraints.
- 71 • We evaluate three GPT-4.1-family API models (OpenAI, 2025) and full 120-item
 72 current-model refreshes for gpt-5.4-mini and gpt-5.5; under gpt-5.5, the contract
 73 improves first-turn global alignment from 81.7% to 98.3% while leaving inspectable
 74 residual failures.
- 75 • We provide diagnostics beyond the headline table: language slices, family-level fail-
 76 ure modes, paired sign-test sensitivity, token-burden analysis, prompt-mechanism
 77 and repair-realism diagnostics, item-consistency analysis, paired judge refreshes,
 78 and launch-ready human/native review protocols with conservative claim gates.

79 We intentionally keep the claim boundary narrow. The pilot is synthetic and stress-oriented,
 80 not a prevalence estimate for all global users. Its purpose is to make a neglected class of
 81 repair costs measurable and reproducible.

82 2 Interaction contracts and metrics

83 **Interaction contract.** For each benchmark item i , we specify a user prompt u_i and a contract
 84 c_i . The contract records expected response language, expected script, required task behavior,
 85 literal spans that must be preserved, forbidden markers or translations, and any register
 86 or locale constraints. Some contract elements are explicit in the prompt, while others are
 87 implicit but recoverable from the workflow. In the Spanish-English editing example above,
 88 the prompt asks for text improvement in Spanish, but the target content is quoted in English.
 89 The contract therefore requires English edited output, not Spanish translation.

First-turn global alignment. For model m and condition s , we define first-turn global alignment as:

$$\text{FTGA}(i, m, s) = \begin{cases} 1 & \text{if the first response satisfies all contract checks,} \\ 0 & \text{otherwise.} \end{cases}$$

Family	User-side ambiguity	Expected contract
Editing preservation	Instruction language differs from quoted content language.	Improve or rewrite the content while preserving its intended language; do not translate unless asked.
Output-language inference	Prompt mixes languages without an explicit “answer in X” instruction.	Infer the expected output language from the pragmatic context.
Quote preservation	The task concerns text around quoted headings or labels.	Preserve required quoted spans and literal data exactly.
Script/register/locale	The user specifies romanization, formal/informal register, or local literal formats.	Obey script, register, and locale constraints while preserving IDs, file names, times, and headings.

Table 1: The four task families in REPROMPTTAX-STRESS-V0.2. The pilot is designed to stress interaction contracts, not to estimate natural prevalence.

The checks cover language, script, preservation, task completion, and register/locale constraints. FTGA is stricter than ordinary task success: a response can complete the requested operation while still failing because it uses the wrong response language, translates a quoted span, or changes a literal file name.

Repair trajectory. If the first response fails, we append up to two standardized repair prompts associated with the item. The repair prompts point to the violated contract class without revealing a reference answer. The **Re-prompt Turn Tax** (RTT) is the minimum number of repair prompts required before success:

$$\text{RTT}(i, m, s) \in \{0, 1, 2, 3\},$$

where 0 denotes first-turn success and 3 denotes unresolved after two repair prompts. We also report Repair@1, Repair@2, and unresolved rate among initially failed trajectories.

Token tax. Let $T_{\text{total}}(i, m, s)$ be provider-reported total tokens consumed by the trajectory through success or exhaustion of the two-repair budget, and let $T_0(i, m, s)$ be the first-attempt token count. We define:

$$\text{TokenTax}(i, m, s) = \frac{T_{\text{total}}(i, m, s)}{T_0(i, m, s)}.$$

Token tax is a repair-burden ratio, not an API-cost estimate. A longer system prompt can reduce repair-normalized token tax while increasing absolute tokens. We therefore report absolute token burden separately in the analysis.

3 Benchmark construction

Motivation from public chat logs. As a lightweight motivation check, we scan the first 20,000 streamed WildChat conversations (Zhao et al., 2024) using predefined repair cues. Among 10,681 multi-turn conversations, 172 contain at least one follow-up cue related to generic repair, preservation, output language, unwanted translation, script, or register/locale. The scan records only aggregate counts and hashed metadata: no raw user or assistant text is written. We use the scan to motivate the failure taxonomy, not to estimate population prevalence.

Stress-pilot design. We created REPROMPTTAX-STRESS-V0.2, a 120-item synthetic benchmark with a balanced 3×4 design: Spanish-English, Hindi-English, and Arabic-English; and four task families with ten items per language-pair/family cell. Table 1 summarizes the families.

The benchmark is synthetic and privacy-safe. A release-hygiene audit confirms 120 unique prompts, zero normalized duplicate prompts, required scoring markers for all rows, zero

113 privacy-marker hits under the audit regexes, and 99 literal preservation spans across the
 114 families that require them. The benchmark also contains known-bad-output notes to clarify
 115 what each item is designed to catch. These checks are not a substitute for native-speaker
 116 validation, but they make the artifact easier to audit and reproduce.

117 4 Experimental protocol

118 **Models.** We evaluate gpt-4.1-nano, gpt-4.1-mini, and gpt-4.1, all accessed through the
 119 API with temperature 0 and a maximum of 256 output tokens. The original paper-facing
 120 experiments were run on June 28, 2026. Because OpenAI’s current model documentation
 121 lists GPT-5.5 as the flagship model and GPT-5.4 mini/nano as lower-cost variants (OpenAI,
 122 2026), we add a current-model refresh on full 120-item gpt-5.4-mini and gpt-5.5 runs
 123 using the same benchmark, conditions, scoring rules, and repair budget.

124 **Prompting conditions.** Each model is run under two system-prompt conditions. The
 125 baseline prompt is “You are a helpful assistant.” The mitigation condition uses a **Global**
 126 **Interaction Contract** prompt that instructs the model to infer instruction and content lan-
 127 guage separately, preserve content language during editing unless translation is requested,
 128 preserve quoted spans and literal data, obey explicit script/register/locale constraints, and
 129 avoid unnecessary explanatory framing.

130 **Repairs.** For each failed first response, we append the item’s first standardized repair
 131 prompt and rescore the next response. If the response still fails, we append a second
 132 standardized repair. The trajectory stops at first success or after two repairs. This fixed
 133 repair budget lets us compare models and prompts under the same user burden. It also
 134 separates first-turn usability from recoverability: some models fail initially but repair quickly,
 135 while others continue to violate literal-preservation constraints.

136 **Scoring.** Automatic scoring combines deterministic checks for exact span preservation
 137 and script with rule-based language and task checks. The scorer emits both a binary FTGA
 138 label and component labels for language, script, preservation, task, and register/locale.
 139 We run scorer-ablation diagnostics by relaxing one component at a time, and deterministic
 140 regression tests cover controlled language, script, preservation, task, and register/locale
 141 failures.

142 **Audit.** We run a blinded gpt-4.1 judge audit on 72 first-turn responses, sampling three
 143 responses from each model-condition-family stratum. The judge sees the user prompt,
 144 expected contract, and assistant response, but not model identity or prompting condition.
 145 This is a scorer sanity check, not a native- or near-native human validation study.

146 5 Results

147 5.1 Main trajectory metrics

148 Table 2 shows measurable baseline Re-prompt Tax for the GPT-4.1-family rows and the
 149 GPT-5.x refresh rows. The Global Interaction Contract improves FTGA for every model,
 150 with GPT-4.1-family gains of +9.2 points for nano, +3.3 for mini, and +16.7 for GPT-4.1;
 151 mean RTT also falls. The strongest GPT-4.1-family model reaches 93.3% FTGA under the
 152 contract, but residual unresolved cases remain in exact-preservation and script/register/
 153 locale rows.

154 Paired sign-test sensitivity supports the strongest effects and bounds mini. For GPT-4.1,
 155 FTGA improves on 20 paired items and worsens none ($p < 0.0001$); mean RTT falls by 0.133
 156 and mean token tax by 0.300. Nano improves on 12 items, worsens on 1 ($p = 0.0034$), and
 157 cuts token tax by 0.35. Mini improves on only 4 items, worsens none ($p = 0.125$), but still
 158 reduces token tax by 0.16.

Model	Condition	FTGA	Mean RTT	Tok. tax	Unres.	R@1	R@2
GPT-4.1 nano	Baseline	67.5	0.47	1.69	5.8	74.4	82.1
GPT-4.1 nano	Contract	76.7	0.33	1.34	4.2	78.6	82.1
GPT-4.1 mini	Baseline	75.8	0.28	1.43	0.8	89.7	96.6
GPT-4.1 mini	Contract	79.2	0.24	1.27	1.7	92.0	92.0
GPT-4.1	Baseline	76.7	0.28	1.43	2.5	89.3	89.3
GPT-4.1	Contract	93.3	0.15	1.13	4.2	37.5	37.5
GPT-5.4 mini	Baseline	80.0	0.25	1.38	2.5	87.5	87.5
GPT-5.4 mini	Contract	85.0	0.25	1.24	5.0	66.7	66.7
GPT-5.5	Baseline	81.7	0.23	1.28	1.7	86.4	90.9
GPT-5.5	Contract	98.3	0.02	1.02	0.0	100.0	100.0

Table 2: Stress-pilot results, including the current-model refresh rows. FTGA, unresolved rate, Repair@1, and Repair@2 are percentages. RTT and token tax are lower-is-better; token tax is total tokens until success or exhaustion divided by first-attempt tokens.

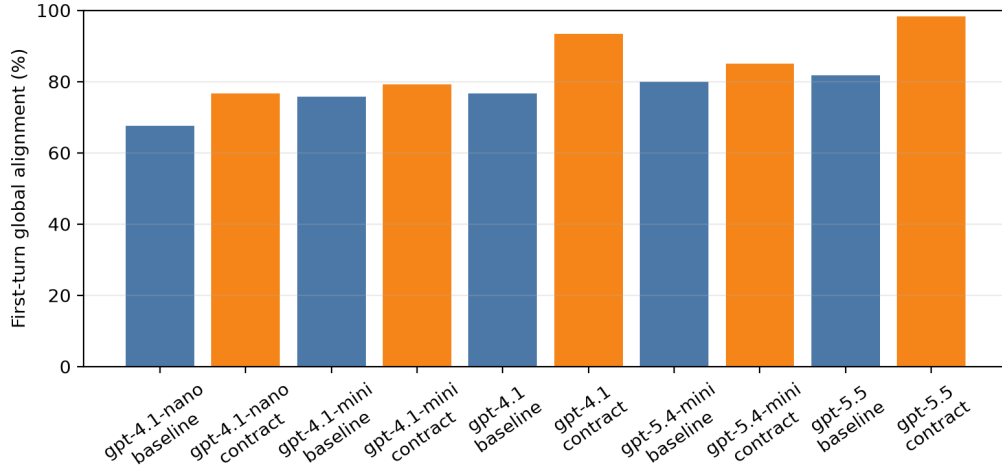


Figure 1: First-turn global alignment by model and condition on the 120-item stress pilot, including the GPT-5.x current-model refresh rows.

159 The current-model refresh strengthens the timeliness claim. On the same 120 items,
 160 gpt-5.4-mini moves from 80.0% to 85.0% FTGA with lower token tax but a higher un-
 161 resolved rate; its +5.0 point bootstrap interval crosses zero at [-0.8, 11.7]. The gpt-5.5 run
 162 is clearer: FTGA rises from 81.7% to 98.3% (+16.7 points, [10.0, 24.2], 20 improved/0 wors-
 163 ened), mean RTT falls from 0.225 to 0.017, and unresolved trajectories fall to 0.0%. It still has
 164 two first-turn failures, both repaired in one turn. A generation-progress probe shows that
 165 GPT-5.x-family baseline failures fall to 46/240 from 96/360 for the GPT-4.1-family baselines.

166 5.2 Failure families

167 The main burden comes from implicit editing preservation, not generic output-language
 168 inference. Table 3 shows aggregate baseline FTGA is 33.3% for editing preservation, with
 169 60/90 first-turn failures, while output-language inference is saturated at 100.0% FTGA in
 170 this pilot. The contract lifts editing-preservation FTGA to 70.0%, lowers mean RTT from
 171 0.63 to 0.30, and moves unresolved rate from 13.3% to 3.3%. Its editing deltas are +63.3,
 172 +13.3, and +33.3 for gpt-4.1, mini, and nano. Component analysis shows why: baseline
 173 editing-preservation rows pass the language check only 33.3% of the time and the script
 174 check 66.7%; the contract raises these to 70.0% and 92.2%, but does not fully solve literal
 175 span preservation. Across all five full-run models, editing preservation contributes 61 of
 176 67 contract fixes and all +61 net first-turn gain. All-five paired sign tests remain positive at

Task family	Baseline FTGA	Contract FTGA	Baseline fail.	Contract fail.
Editing preservation	33.3	70.0	60/90	27/90
Output-language inference	100.0	100.0	0/90	0/90
Quote preservation	90.0	91.1	9/90	8/90
Script/register/locale	70.0	71.1	27/90	26/90

Table 3: Family-level first-turn alignment aggregated across the three models. Editing preservation is the dominant source of baseline burden; output-language inference is saturated in this pilot.

model-item (67 fixes vs. 6 regressions, $p < 0.0001$) and item-cluster units (24 net-positive vs. 5 net-negative items, $p = 0.0005$). Resampling prompt items, the clustered bootstrap interval is +5.8 to +15.0 points.

5.3 Language slices and item consistency

The aggregate gain is uneven across language pairs: averaged over models, FTGA increases by +20.0 points on Arabic-English, +9.2 on Spanish-English, and +0.0 on Hindi-English. This is not a population claim; in this stress pilot, Hindi-English leaves little room to improve because baseline FTGA is already high. Item-consistency analysis separates systematic hard items from one-model failures: baseline prompting leaves 40/120 items failing for at least one model and 27/120 failing for all three, while the contract reduces all-model failures to 8/120 and the number of failing models on 22/120 items.

5.4 Repair dynamics and token burden

Repair trajectories distinguish first-turn usability from recoverability. For GPT-4.1, the contract changes 92 first-turn successes and 28 initial failures to 112 successes and 8 failures; for nano, successes rise from 81 to 92 and unresolved trajectories fall from 7 to 5. Token tax decreases for all three models, but absolute token burden increases because the contract prompt is longer. For GPT-4.1, total tokens rise from 132.7 to 256.3 even as post-first turn repair tokens decrease from 51.2 to 32.9, so token tax is normalized repair burden rather than cost savings. A 24-item repair-realism diagnostic further shows repair wording sensitivity: saved standard and terse repairs recover 24/24 editing-preservation failures, while frustrated repair recovers 17/24 and explicit source-language contract repair only 5/24. We treat this as interaction-realism evidence, not as a user study.

5.5 Prompt controls and mechanism evidence

Single-model prompt diagnostics on gpt-4.1-nano clarify mechanism without widening the main claim. A generic-helpfulness prompt improves FTGA from 67.5% to 75.0%; the full contract reaches 76.7%; and a narrower content-preservation prompt reaches 80.0% FTGA, 0.27 mean RTT, and $1.28\times$ token tax. Thus the paper-facing intervention remains the full contract, but content-language and literal-preservation rules appear to drive much of the mitigation. On current models, the narrower prompt is a near tie: 85.8% versus 85.0% FTGA on gpt-5.4-mini, and 99.2% versus 98.3% on gpt-5.5. We treat preservation scaffolding as mechanism evidence, not as a uniformly better prompt.

5.6 Scorer sensitivity and audit

Scorer-ablation diagnostics show that the headline trend is not caused by one fragile rule. Relaxing language alone would move aggregate baseline FTGA from 73.3% to 76.4%; relaxing preservation alone to 78.9%; relaxing task would not change it. Under the contract, relaxing language would move FTGA from 83.1% to 83.3%, and relaxing preservation to 88.3%. The task-useful diagnostic further shows hidden repair burden: 31/96 baseline first-turn failures still pass the task component, and the stricter task-plus-preservation slice falls from 11 to 1 under the contract.

The blinded judge audit agrees with the corrected automatic scorer on 71/72 sampled pass/fail labels (98.6%; Wilson 95% CI 92.5–99.8). A paired gpt-5.5 judge refresh on the same 72 rows agrees with the automatic scorer on 70/72 labels and with the original gpt-4.1 judge on 69/72 labels. Component agreement is lower for preservation and register/locale, reinforcing the claim boundary: LLM judges are useful sanity checks, but native/near-native validation remains necessary before stronger cultural or register claims.

6 Discussion

Re-prompt Tax changes the unit of analysis from a single answer to a repair trajectory. This matters because a first answer can be semantically plausible but interactionally wrong. For global users, the cost of that error includes not only extra tokens, but also the cognitive work of diagnosing the failure, writing a corrective prompt, and verifying the repair. These costs are easy to miss when benchmarks ask only whether a final response is acceptable.

The strongest empirical finding is that implicit content-language preservation is a measurable bottleneck. Models often follow the language of the instruction rather than the language of the quoted content to be edited. A simple contract prompt reduces this failure, but does not eliminate exact-preservation errors. This pattern suggests that multilingual system prompts should separate *instruction language*, *content language*, and *response language* as distinct variables rather than treating the prompt as a monolithic language signal.

The results also show why mitigation claims should be bounded. The full Global Interaction Contract improves all three models, but it should be read as a pre-registered intervention, not as the best possible prompt: a narrower content-preservation prompt outperforms it on the nano diagnostic and near-ties it on current gpt-5.4-mini and gpt-5.5 diagnostics. This suggests that more specific scaffolds may be better than broad multilingual policy text for some workflows, but the current-model gaps are too small to claim prompt dominance. Future systems could infer a lightweight interaction contract from the user’s prompt and attach only the relevant constraints, reducing both absolute prompt length and over-instruction risk.

7 Related work

Multilingual and cultural evaluation. Multilingual benchmark work has expanded language coverage and highlighted large performance gaps across languages. MMLU-ProX evaluates advanced reasoning across typologically diverse languages (Xuan et al., 2025), BenchMAX proposes a broad multilingual evaluation suite (Huang et al., 2025b), and OMGEval evaluates open-ended multilingual generation with localized questions (Liu et al., 2024). Global MMLU shows that simply translating English benchmarks can preserve Western-centric cultural assumptions and distort model rankings (Singh et al., 2024). Cultural benchmarks such as CDEval and MCEval argue that model behavior should be evaluated through culturally specific and cross-cultural lenses (Wang et al., 2023; Huang et al., 2025a). Our work is smaller and narrower: it does not attempt broad cultural coverage, but instead measures the interactional cost of recovery after violating multilingual interaction contracts.

Instruction following and code-switching. IFEval introduced verifiable instruction-following checks for LLMs (Zhou et al., 2023); M-IFEval, XIFBench, and Marco-Bench-MIF extend this concern to multilingual instruction following with language-specific and localized constraints (Dussolle et al., 2025; Li et al., 2025b; Zeng et al., 2025). OLA is the closest prior work to ours: it studies output-language alignment in code-switched interactions and shows that models often infer the wrong response language (Oh et al., 2026). CodeMixQA and related work further show that code-switched reasoning and generation remain challenging (Winata et al., 2026). Re-prompt Tax includes output-language alignment, but adds repair trajectories and a broader contract containing script, preservation, register, and locale constraints, and standardized repairs are needed to compare those failures as trajectories rather than isolated first-turn answers.

Multi-turn dialogue, real-world chat logs, and judges. MT-Bench and Chatbot Arena evaluate conversational quality and preference using multi-turn prompts and LLM or human judgments (Zheng et al., 2023b). WildChat and LMSYS-Chat-1M provide large-scale public records of real LLM interactions (Zhao et al., 2024; Zheng et al., 2023a). We use WildChat only for a bounded aggregate cue scan and release synthetic stress items instead of raw user conversations. Recent multi-turn instruction-following benchmarks such as StructFlowBench and MultiChallenge evaluate structural dependencies, context allocation, and realistic dialogue challenges (Li et al., 2025a; Sirdeshmukh et al., 2025). Re-prompt Tax is more controlled: it fixes the failure-triggered repair prompt and measures recovery cost after a multilingual contract violation. Our judge audit is related to LLM-as-a-judge methodology, but multilingual judging itself is known to be inconsistent across languages (Fu & Liu, 2025). We therefore treat the judge audit as a sanity check rather than as final human validation.

Cost and accessibility. Prior work on multilingual tokenization shows that language users can face unequal token costs and utility when using commercial APIs (Ahia et al., 2023; Petrov et al., 2023). Re-prompt Tax is complementary: instead of asking whether the same content costs more to encode in different languages, we ask how many extra turns and tokens are consumed when the model violates a multilingual interaction contract.

8 Limitations and ethics

This is a pilot benchmark, not a representative sample of multilingual or code-switched use. It covers only three language pairs and uses synthetic prompts. The English content embedded in the items reflects a common editing workflow, but many global workflows involve non-English target content, non-Latin scripts, dialectal variation, or richer social-register distinctions that are not covered here.

The automatic scorer is intentionally conservative and strongest for exact span-preservation and script checks. It is weaker for nuanced register and cultural appropriateness. Two blinded LLM-judge audits support most sampled pass/fail labels, but they also disagree on a small number of borderline editing, register, and concision cases. They are not a substitute for native/near-native human validation; native-speaker audit remains necessary before stronger final claims. Any release or final workshop version should clearly label the human audit status and avoid claims that the benchmark validates cultural appropriateness for all included communities.

The WildChat cue scan uses only aggregate counts and hashed metadata. We do not release raw user logs or quote private conversation text. The scan is useful for motivating the taxonomy, but regex cue hits are not reliable prevalence estimates. The benchmark items are synthetic and should be used to stress-test interaction failure modes, not to characterize all speakers. Any release should include a benchmark card explaining intended use, non-representativeness, language-pair coverage, and known scorer limitations.

Finally, token tax is a normalized repair-burden metric, not a direct economic claim. In our experiments, the longer contract prompt reduces repair-normalized token tax but increases absolute token use. Reporting both numbers is important for users whose constraints are monetary cost, latency, or context-window budget.

9 Conclusion

We introduced Re-prompt Tax as a way to measure hidden interaction costs for multilingual and code-switched LLM users. On a 120-item synthetic stress pilot, three GPT-4.1-family models exhibit measurable first-turn contract failures, especially on implicit editing-preservation tasks. A lightweight Global Interaction Contract prompt improves first-turn global alignment and reduces repair burden, and a full current-model refresh shows the same pressure on gpt-5.5: first-turn alignment improves from 81.7% to 98.3% but is not perfect. Follow-up prompt-mechanism diagnostics indicate that narrower content-preservation

scaffolding captures much of the effect, but the full contract remains the paper-facing intervention. Residual exact-preservation and script/register/locale failures remain. The broader takeaway is methodological: global evaluation should measure not only whether a model can eventually produce a correct answer, but also how much repair work users must perform to make the answer usable.

References

- Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David R. Mortensen, Noah A. Smith, and Yulia Tsvetkov. Do all languages cost the same? tokenization in the era of commercial language models, 2023. URL <https://arxiv.org/abs/2305.13707>.
- Antoine Dussolle, Andrea Cardeña Díaz, Shota Sato, and Peter Devine. M-IFEval: A multilingual instruction-following evaluation benchmark, 2025. URL <https://arxiv.org/abs/2502.04688>.
- Xiyan Fu and Wei Liu. How reliable is multilingual LLM-as-a-judge?, 2025. URL <https://arxiv.org/abs/2505.12201>.
- GenAI4World Organizers. Generative ai for the world: Workshop on globalizing tasks, evaluations, and systems at colm 2026, call for papers. <https://sites.google.com/view/genai4world/call-for-papers>, 2026. Accessed 2026-06-28.
- Shulin Huang, Linyi Yang, and Yue Zhang. MCEval: A dynamic framework for fair multilingual cultural evaluation of LLMs, 2025a. URL <https://arxiv.org/abs/2507.09701>.
- Xu Huang, Wenhao Zhu, Hanxu Hu, Conghui He, Lei Li, Shujian Huang, and Fei Yuan. BenchMAX: A comprehensive multilingual evaluation suite for large language models, 2025b. URL <https://arxiv.org/abs/2502.07346>.
- Jinnan Li, Jinzhe Li, Yue Wang, Yi Chang, and Yuan Wu. StructFlowBench: A structured flow benchmark for multi-turn instruction following, 2025a. URL <https://arxiv.org/abs/2502.14494>.
- Zhenyu Li, Kehai Chen, Yunfei Long, Xuefeng Bai, Yaoyin Zhang, Xuchen Wei, Juntao Li, and Min Zhang. XIFBench: Evaluating large language models on multilingual instruction following, 2025b. URL <https://arxiv.org/abs/2503.07539>.
- Yang Liu, Meng Xu, Shuo Wang, Liner Yang, Haoyu Wang, Zhenghao Liu, Cunliang Kong, Yun Chen, Yang Liu, Maosong Sun, and Erhong Yang. OMGEval: An open multilingual generative evaluation benchmark for large language models, 2024. URL <https://arxiv.org/abs/2402.13524>.
- Juhyun Oh, Haneul Yoo, Faiz Ghifari Haznitrama, and Alice Oh. OLA: Output language alignment in code-switched LLM interactions, 2026. URL <https://arxiv.org/abs/2601.03589>.
- OpenAI. Introducing gpt-4.1 in the api. <https://openai.com/index/gpt-4-1/>, 2025. Accessed 2026-06-28.
- OpenAI. Models: Openai api documentation. <https://platform.openai.com/docs/models>, 2026. Accessed 2026-06-28.
- Aleksandar Petrov, Emanuele La Malfa, Philip H. S. Torr, and Adel Bibi. Language model tokenizers introduce unfairness between languages, 2023. URL <https://arxiv.org/abs/2305.15425>.
- Shivalika Singh, Angelika Romanou, Clémentine Fourier, David I. Adelani, Jian Gang Ngui, Daniel Vila-Suero, Peerat Limkonchotiawat, Kelly Marchisio, Wei Qi Leong, Yosephine Susanto, et al. Global MMLU: Understanding and addressing cultural and linguistic biases in multilingual evaluation, 2024. URL <https://arxiv.org/abs/2412.03304>.

- 364 Ved Sirdeshmukh, Kaustubh Deshpande, Johannes Mols, Lifeng Jin, Ed-Yeremai Cardona,
365 Dean Lee, Jeremy Kritz, Willow Primack, Summer Yue, and Chen Xing. MultiChallenge:
366 A realistic multi-turn conversation evaluation benchmark challenging to frontier llms,
367 2025. URL <https://arxiv.org/abs/2501.17399>.
- 368 Yuhang Wang, Yanxu Zhu, Chao Kong, Shuyu Wei, Xiaoyuan Yi, Xing Xie, and Jitao Sang.
369 CDEval: A benchmark for measuring the cultural dimensions of large language models,
370 2023. URL <https://arxiv.org/abs/2311.16421>.
- 371 Genta Indra Winata, David Anugraha, Patrick Amadeus Irawan, Anirban Das, Haneul
372 Yoo, Paresch Dashore, Shreyas Kulkarni, Ruochen Zhang, Haruki Sakajo, et al. Can large
373 language models understand, reason about, and generate code-switched text?, 2026. URL
374 <https://arxiv.org/abs/2601.07153>.
- 375 Weihao Xuan, Rui Yang, Heli Qi, Qingcheng Zeng, Yunze Xiao, et al. MMLU-ProX: A
376 multilingual benchmark for advanced large language model evaluation, 2025. URL
377 <https://arxiv.org/abs/2503.10497>.
- 378 Bo Zeng, Chenyang Lyu, Sinuo Liu, Mingyan Zeng, Minghao Wu, Xuanfan Ni, Tianqi Shi,
379 Yu Zhao, Yefeng Liu, Chenyu Zhu, et al. Marco-Bench-MIF: On multilingual instruction-
380 following capability of large language models, 2025. URL <https://arxiv.org/abs/2507.11882>.
- 382 Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, and Yuntian Deng. Wild-
383 Chat: 1m chatgpt interaction logs in the wild, 2024. URL <https://arxiv.org/abs/2405.01470>.
- 385 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu,
386 Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric P. Xing, Joseph E. Gonzalez, Ion Stoica, and
387 Hao Zhang. LMSYS-Chat-1M: A large-scale real-world LLM conversation dataset, 2023a.
388 URL <https://arxiv.org/abs/2309.11998>.
- 389 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, et al. Judging
390 LLM-as-a-judge with MT-Bench and chatbot arena, 2023b. URL <https://arxiv.org/abs/2306.05685>.
- 392 Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny
393 Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023.
394 URL <https://arxiv.org/abs/2311.07911>.